

Self-Check Sheet

Table of contents

Week 1: Kickstart with R	1
Week 2: Vectors and Matrices	2
Week 3: Data Frames and Lists	3
Week 4: Operators, Conditions, and Functions	3
Week 5: ggplot2 (part 1)	4
Week6: Revisiting the Essentials	4
Week7: Wrangling Data with dplyr and tidyr	5
Week 8: ggplot2 (part 2) & Custom Functions	6
Week 9: Reproducible Analysis with Quarto	6

Notes:

- Elements appear in the chronological order we saw them in class.
- Elements marked with * were mentioned in class but are **not required** to be mastered.

Week 1: Kickstart with R

```
sqrt()
ls()
library()
mean()
q()
?
getwd()
file.exists()
readr::read_csv()
head()
View()
readr::write_csv()
```

dim()
mode()
sd()
summary()
*hist()
*plot()
log2()

*text()
*abline()

Week 2: Vectors and Matrices

help()
NA
is.numeric()
is.double()
is.integer()
as.numeric()
typeof()
is.character()
as.character()
is.logical()

c() :
seq()
paste0()
rep()
is.vector()
length()
names()
str()
tail()
is.na()
sum()
median()

matrix()
rbind()
cbind()
nrow()
ncol()
rownames()

```
colnames()
is.matrix()
rowSums()
colSums()
rowMeans()
colMeans()

*set.seed()
round()
*rnorm()
shapiro.test()
*curve()
*dnorm()
```

Week 3: Data Frames and Lists

```
data.frame()
is.data.frame()
unlist()
NULL
factor()
merge()

list()
names()

read.table()
unique()
wilcox.test()
*boxplot()
```

Week 4: Operators, Conditions, and Functions

```
<-
=
[]
$
[[]]
+, -, *, /, ^
<, <=, >, >=, ==, !=
!, &, |
%in%
any()
```

```
all()
which()
if()
if () {} else {}
ifelse()
print()

abs()
table()
```

Week 5: ggplot2 (part 1)

```
install.packages()
remove.packages()

ggplot()
aes()
geom_point()
labs()
geom_smooth()
geom_boxplot()
geom_violin()
geom_histogram()
geom_bar()
theme_grey()
theme_bw()
theme_classic()
theme_bw()
theme()
ggsave()

as.data.frame()
theme_minimal()
*element_text()
```

Week6: Revisiting the Essentials

```
read.csv()
write.table()
write.csv()
readxl::read_xlsx()
readxl::read_xls()
writexl::write_xlsx()
```

```
saveRDS()  
readRDS()  
save()  
t.test()
```

Week7: Wrangling Data with dplyr and tidyr

```
|>  
*%>%  
  
tibble()  
as_tibble()  
  
pivot_longer()  
pivot_wider()  
drop_na()  
  
select()  
starts_with()  
ends_with()  
contains()  
matches()  
mutate()  
rename()  
filter()  
arrange()  
desc()  
count()  
summarise()  
group_by()  
ungroup()  
  
*pull()  
*if_else()  
*slice()  
*slice_head()  
*slice_tail()  
  
*readr::read_csv2()  
*extract()  
*GGally::ggpairs()  
  
cor()
```

Week 8: ggplot2 (part 2) & Custom Functions

```
theme_set()
theme_update()
scale_color_manual()
scale_fill_manual()
scale_shape_manual()
scale_x_continuous()
scale_y_continuous()
xlim()
ylim()
geom_hline()
geom_vline()
facet_wrap()
facet_grid()
geom_label()
*ggrepel::geom_label_repel()
*ggpubr::stat_compare_means()

theme_minimal()
*everything()
*sub()
*rank()
*position_jitter()
paste()

*ggpubr::stat_pvalue_manual()
geom_col()

*patchwork::wrap_plots()
*patchwork::plot_annotation()
```

Week 9: Reproducible Analysis with Quarto

```
# H1
## H2
### H3
**Bold**
*Italic*
`code`
`r nrow(iris)`
Superscript1
```

```
Subscript~2~  
![] (path/to/image.png)  
- unordered list  
  - unordered sub-list  
1. ordered list
```

```
left_join()  
theme_light()  
  
aov()  
*TukeyHSD()  
  
*bartlett.test()  
*kruskal.test()
```